

Macroeconomics

ECON 101

Fiscal Policy

(Chapter 12)

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Chapter Outline

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What is Fiscal Policy

12.1 Learning Objective

Define fiscal policy.

Fiscal policy refers to changes in *federal* taxes and purchases that are intended to achieve macroeconomic policy objectives.

(Provincial and territorial taxes and spending are not generally aimed at affecting *national-level* objectives.)

Some forms of government spending and taxes automatically increase or decrease along with the business cycle; these are **automatic stabilizers**.

Example: *Employment insurance payments are larger during a recession.*

Discretionary fiscal policy, on the other hand, refers to intentional actions the government takes to change spending or taxes.

The Effects of Fiscal Policy on Real GDP and the Price Level

12.2 Learning Objective

Explain how fiscal policy affects aggregate demand and how the government can use fiscal policy to stabilize the economy.

The government carry out fiscal policy through:

- Changes in government purchases
- Changes in taxes

A change in government purchases **directly** affects aggregate demand.

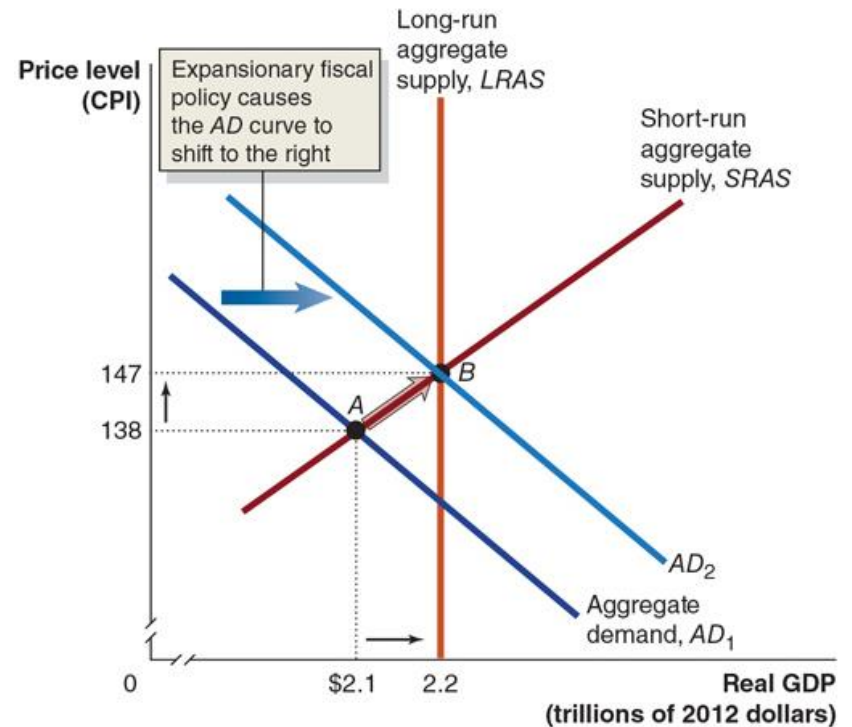
A change in taxes changes income; this in turn affects consumption, and so it has an **indirect** effect on aggregate demand.

Figure 12.7 Fiscal Policy (1 of 2)

Expansionary fiscal policy involves increasing government purchases or decreasing taxes.

- Increasing government purchases directly increases aggregate demand.
- Decreasing taxes indirectly affects aggregate demand by increasing disposable income, and hence consumption spending.

*If the government believes real GDP will be **below potential GDP**, it can enact **expansionary fiscal policy** in an attempt to restore long-run equilibrium—decreasing unemployment.*



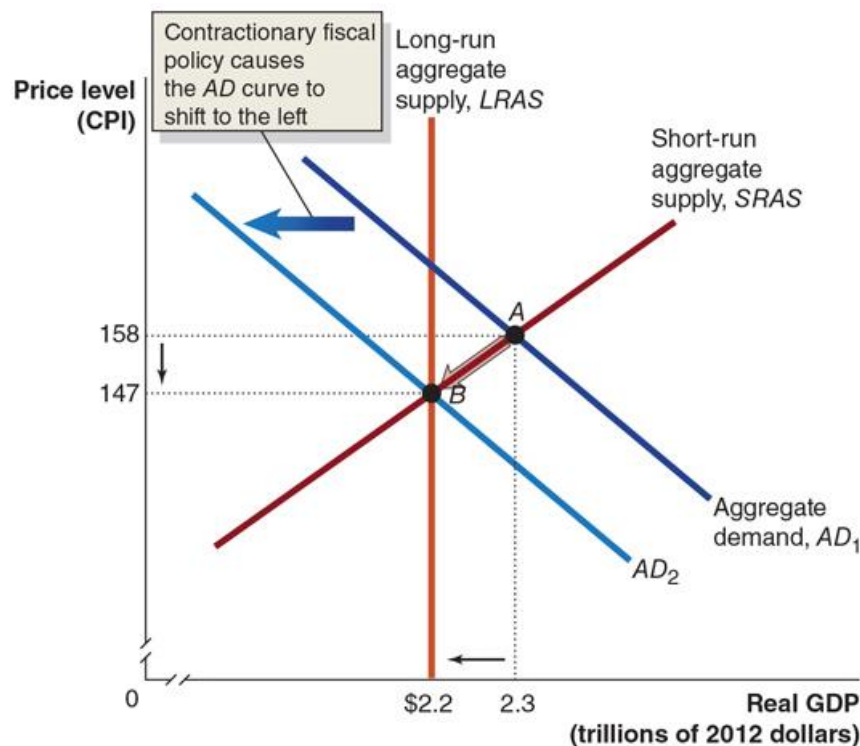
(a) Expansionary fiscal policy
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Figure 12.7 Fiscal Policy (2 of 2)

Contractionary fiscal policy involves *decreasing* government purchases or *increasing* taxes.

- This works just like expansionary fiscal policy, only in reverse.

If the government believes real GDP will be **above potential GDP**, it can enact **contractionary fiscal policy** in an attempt to restore long-run equilibrium—decreasing inflation.



(b) Contractionary fiscal policy

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Table 12.1 Countercyclical Fiscal Policy

Problem	Type of Policy Required	Actions by Congress and the President	Result
Recession	Expansionary	Increase government purchases or cut taxes	Real GDP and the price level rise.
Rising inflation	Contractionary	Decrease government purchases or raise taxes	Real GDP and the price level fall.

Bear in mind that:

- The effects described assume **ceteris paribus**: everything else is staying the same, including **monetary policy**.
- Contractionary fiscal policy is not really causing prices to fall; it's causing inflation to be lower **than it otherwise would have been**.

The Government Purchases and Tax Multipliers

12.4 Learning Objective

Explain how the government purchases and tax multipliers work.

If the government increases its spending on goods and services, then aggregate demand increases immediately. This is the **autonomous** increase in aggregate demand.

But then people receive this increased spending as increased income, increasing their consumption spending accordingly. This is the **induced** increase in aggregate demand.

The series of induced increases in consumption spending that results from the initial increase in autonomous expenditures is known as the **multiplier effect**.

Figure 12.10 The Multiplier Effect and Aggregate Demand

Predicting the eventual (multiplied) effect of a change in autonomous expenditures requires knowing how big the multiplier is.

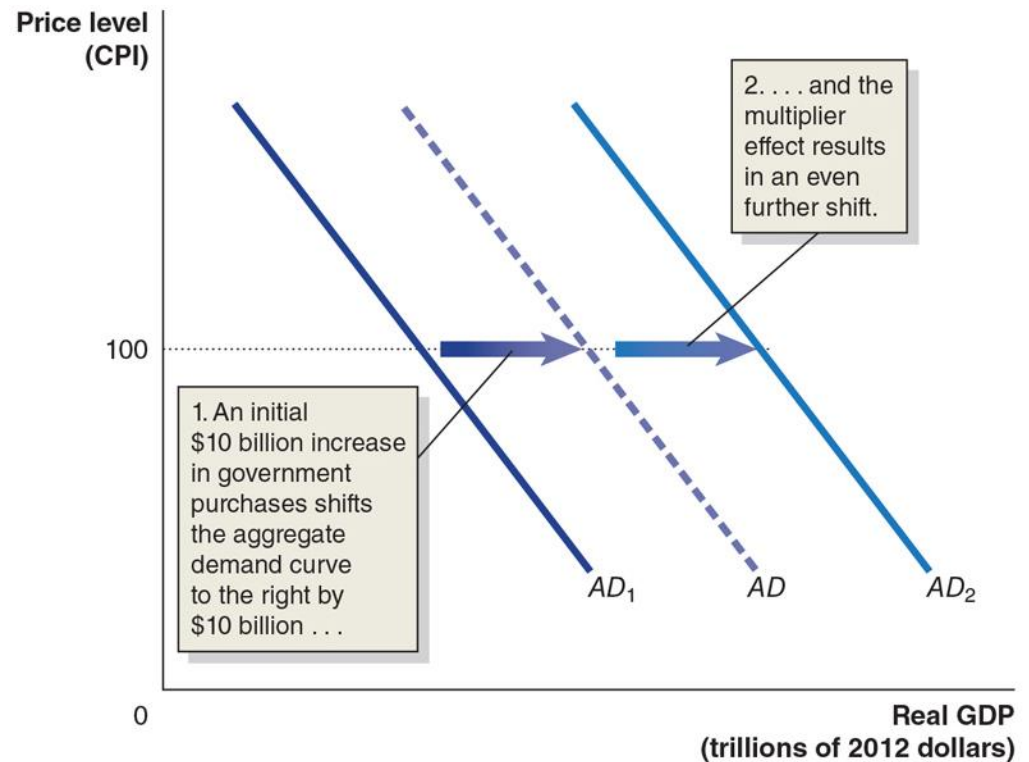
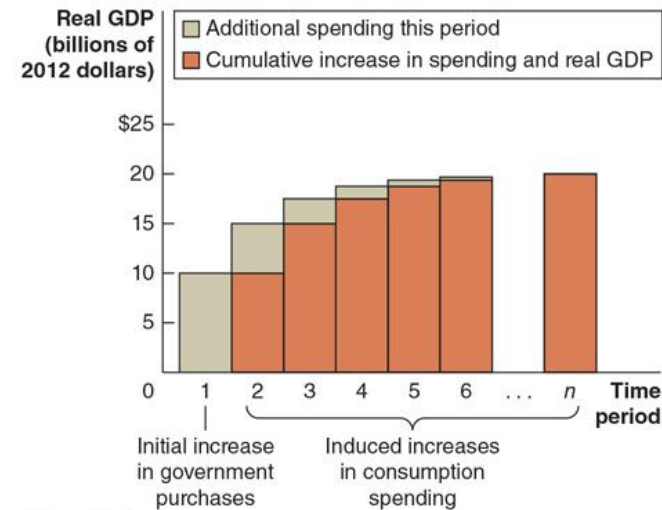


Figure 12.11 The Multiplier Effect of an Increase in Government Purchases

Time Period	Additional Spending This Period	Cumulative Increase in Spending and Real GDP
1	\$10 billion in government purchases	\$10 billion
2	\$5 billion in consumption spending	\$15 billion
3	\$2.5 billion in consumption spending	\$17.5 billion
4	\$1.25 billion in consumption spending	\$18.75 billion
5	\$0.625 billion in consumption spending	\$19.375 billion
6	\$0.3125 billion in consumption spending	\$19.6875 billion
⋮	⋮	⋮
n	0	\$20 billion



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Suppose each increase in spending induces an increase in consumption spending by one-half of the increase in income from the previous period, on domestically produced goods and services.

- Over time, a \$10 billion increase in government purchases will result in an additional \$10 billion in induced consumption spending.

Multipliers for Government Purchases and Taxes

We can describe the total effect of a change (increase *or* decrease) in government purchases or taxes by measuring the change in equilibrium real GDP.

$$\text{Government purchases multiplier} = \frac{\text{Change in equilibrium real GDP}}{\text{Change in government purchases}}$$

$$\text{Tax multiplier} = \frac{\text{Change in equilibrium real GDP}}{\text{Change in taxes}}$$

The tax multiplier will be a negative number: an increase in taxes will *decrease* equilibrium real GDP, and vice versa.

We expect the tax multiplier to be *smaller* (in absolute value) than the government purchases multiplier.

- Why? A \$10 billion increase in purchases initially increases spending by \$10 billion, but a \$10 billion tax cut is partially spent and partially saved.

The Effect of Changes in Tax Rates

The tax multiplier applies to changes in the *amount* of taxes, without changes in *tax rates*.

Decreases in tax rates have a slightly different effect:

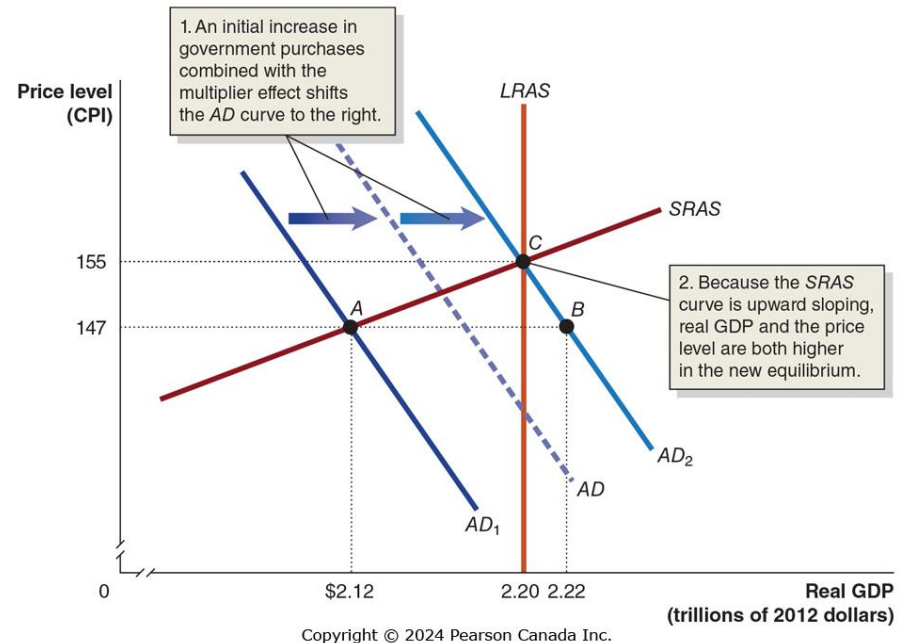
1. Increasing the disposable income of households, leading them to increase their consumption spending.
2. Increasing the size of the multiplier effect, since more of any increase in income becomes disposable income.

Figure 12.12 The Multiplier Effect and Aggregate Supply

An increase in aggregate demand will not only result in real GDP rising; it will also result in a price level increase, because the short-run aggregate supply (SRAS) curve is upward-sloping.

Suppose that between the autonomous and induced effects, fiscal policy causes aggregate demand to increase by \$100 billion.

- The resulting real GDP increase is smaller—only \$100 billion (from \$2.12 trillion to \$2.22 trillion). The price level also rises.



The Multipliers Work in Both Directions

An increase in government purchases and a cut in taxes have a positive multiplier effect.

A decrease in government purchases and an increase in taxes have a *negative* multiplier effect.

Example: a reduction in government spending on infrastructure initially affects contractors, but then it would spread to suppliers and employees of those contractors, and then to other firms and workers.

The Limits of Fiscal Policy as a Stimulus

12.5 Learning Objective

Discuss the difficulties that can arise in implementing fiscal policy.

For several reasons, fiscal policy may be even less effective than monetary policy at countercyclical stabilization:

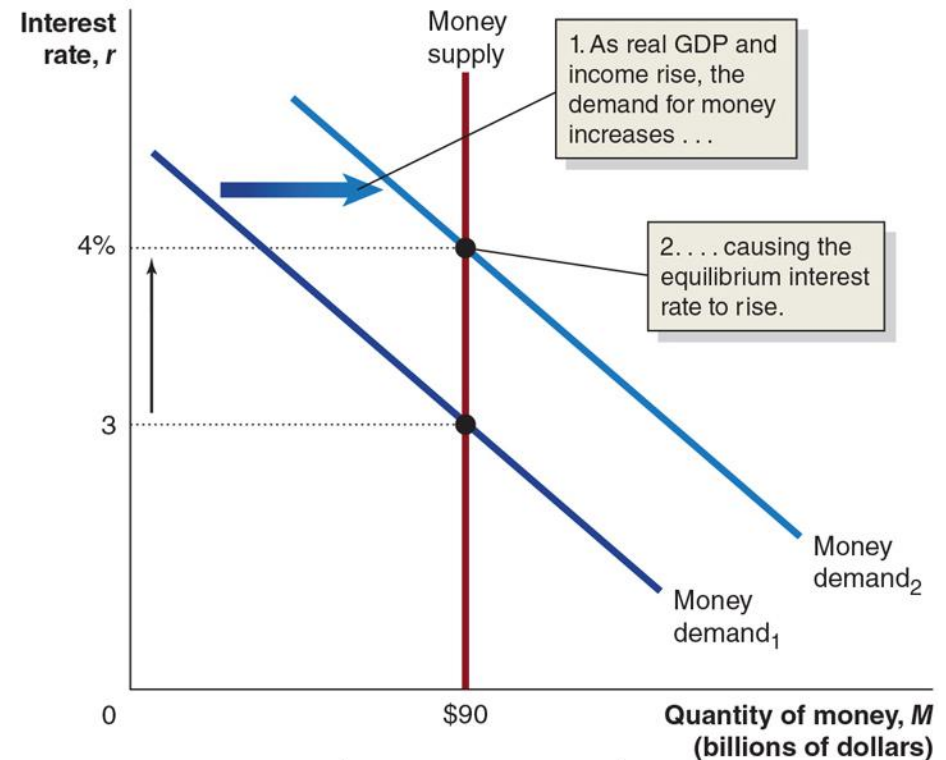
- **Timing** of fiscal policy is harder, due to:
 - **Legislative delay:** *Parliament needs to agree on the actions*
 - **Implementation delay:** *Large spending projects may take months or even years to begin, even after approval.*
- Government spending might **crowd out** private investment.

Crowding out: A decline in private expenditures as a result of an increase in government purchases

Figure 12.13 An Expansionary Fiscal Policy Increases Interest Rates

A temporary increase in government purchases will cause the demand for money, and hence the interest rate, to rise.

But with the higher interest rate, consumption, investment, and net exports all fall.



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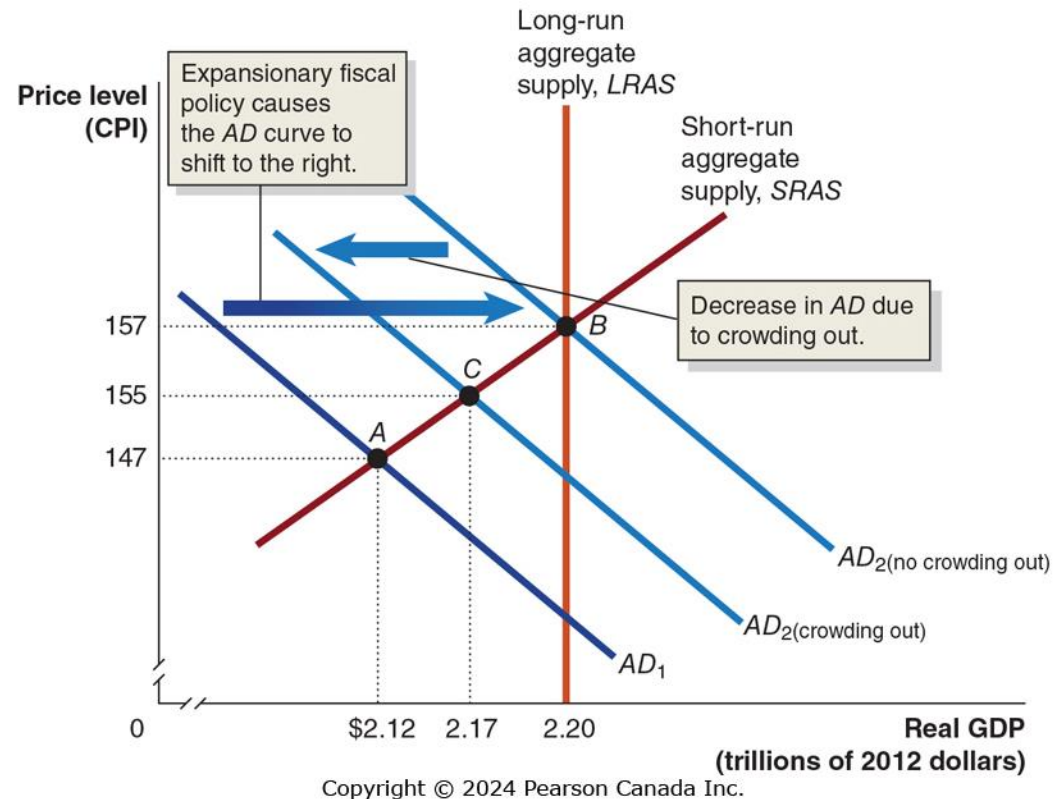
Figure 12.14 The Effect of Crowding Out in the Short Run

A temporary increase in government purchases will cause the demand for money, and hence the interest rate, to rise.

- With the higher interest rate, consumption, investment, and net exports all fall.

So the initial increase in spending...

... is partially offset because of crowding out.



Crowding Out in the Long Run

In the long run, the increase in government purchases will have no effect on real GDP; the reduction in consumption, investment, and net exports will exactly offset the increase in government purchases.

- Why? Because *in the long run, the economy returns to potential GDP*, even without the government's intervention.

Deficits, Surpluses, and Federal Government Debt

12.6 Learning Objective

Define federal budget deficit and federal government debt, and explain how the federal budget can serve as an automatic stabilizer.

A **budget deficit** is the situation in which the government's expenditures are greater than its tax revenue.

A **budget surplus** is the situation in which the government's expenditures are less than its tax revenue.

Do you know whether the federal government is running a budget deficit or a budget surplus currently?

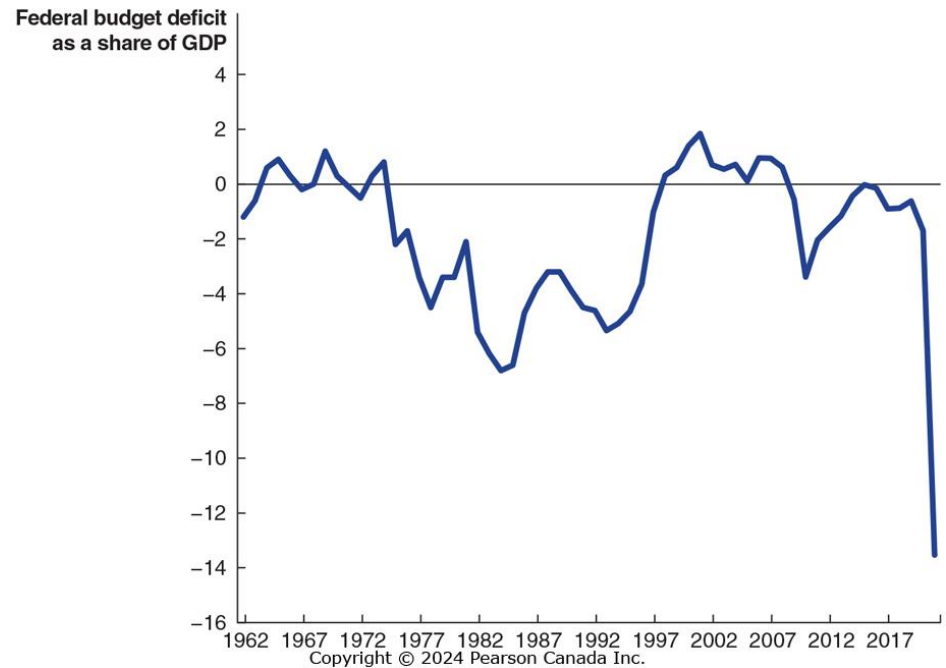
Figure 12.15 The Federal Budget Deficit, 1962–2021

The federal government has run deficits for most years since 1962.

Deficits are larger during recession than expansions, but not always.

Typical during recessions, as tax receipts fall, and **automatic stabilizers**, like increases in transfer payments (unemployment insurance, welfare, etc.), take effect.

These automatic stabilizers are important for limiting the severity of a recession; many economists believe that the Great Depression of the 1930s was more severe because most of these automatic stabilizers did not exist then.



SOURCE: Government of Canada, Department of Finance, “Fiscal Reference Tables for 2021,” <https://www.canada.ca/en/departement-finance/services/publications/fiscal-reference-tables/2021.html>, and for 1996, <https://www.canada.ca/content/dam/fin/migration/frt-trf/pdf/frt96e.pdf>.

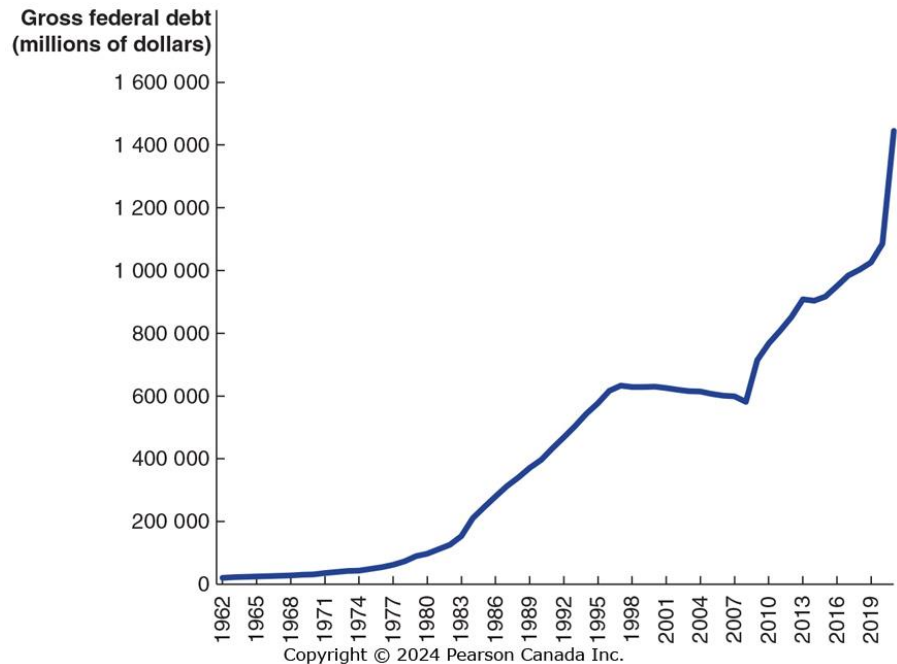
Figure 12.16 The Federal Government Debt, 1962–2021

When the federal government runs a budget deficit, it finances its activities by selling securities (bonds).

The total value of those securities outstanding is known as the *federal government debt*, or the *national debt*.

With the exception of 2000 to 2008, Canada's debt has been growing the entire period of time.

The debt stabilized again in 2013–2014 but began rising in 2015. In 2020 to 2022, the federal debt rose dramatically as government revenue initially fell and expenditures rose rapidly.



SOURCE: Government of Canada, Department of Finance, “Fiscal Reference Table for 2021,” <https://www.canada.ca/en/department-finance/services/publications/fiscal-reference-tables/2021.html>, and for 1996, <https://www.canada.ca/content/dam/fin/migration/frt-trf/pdf/frt96e.pdf>.

Is Government Debt a Problem?

For now, the federal government is at no serious risk of *defaulting* on its debt, because:

- The interest rate it can borrow money at is *very* low, and
- The size of the interest payments on the debt is low relative to the size of the federal budget

In the long run, a debt that increases in size relative to GDP can pose a problem—potentially crowding out investment, which is a key component of long term growth.

- This problem is reduced if the government debt was incurred to finance *infrastructure*, education, or research and development; these serve as a long term investment for the economy.

The Effects of Fiscal Policy in the Long Run

12.7 Learning Objective

Discuss the effects of fiscal policy in the long run.

The fiscal policy we have concentrated on so far was intended to address short-run goals of stabilizing the economy.

- But other fiscal policy actions are intended to have long-run impacts on potential GDP—i.e. on **aggregate supply**, rather than **aggregate demand**.
- Hence these actions are often referred to as **supply-side economics**.

Most of these policies are based on changing taxes in order to increase incentives to work, save, invest, and start a business.

The Long-Run Effects of Tax Policy

Most taxes are assessed as a percentage of some economic activity, like individual income, corporate income, or capital gains.

- When an individual decides how much to work, he bases the decision on how much an hour of work will increase his ability to consume goods and services—the *post-tax wage*.
- When a firm decides how many people to employ, it considers how much it has to pay in total for each worker: the *pre-tax wage*.

The difference between the pretax and post-tax return to an economic activity is known as the **tax wedge**. It applies to the *marginal tax rate* which is the fraction of each additional dollar of income paid in taxes.

- A large tax wedge *distorts the incentives* of individuals and firms to take part in economic activities, generally resulting in lower levels of economic activity—lower real GDP.

Tax Rates Matter

Tax rates matter because the larger they are, the larger will be the behavioral response to the tax:

Individual income tax

- Affects labour supply decisions and the returns to entrepreneurship

Corporate income tax

- Affects the incentives of firms to engage in investment

Tax on dividends and capital gains

- Affects the supply of loanable funds from households to firms, and hence the real interest rate.
- Also affects the way firms disburse profits.

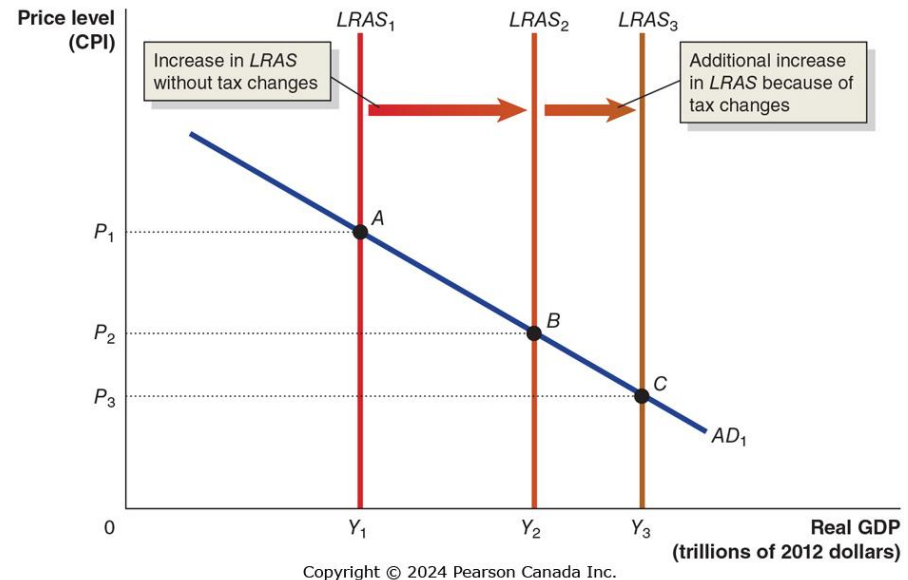
Figure 12.17 The Supply-Side Effects of a Tax Change

Tax reform has the potential to increase real GDP in the long run beyond the increases that would occur without tax changes.

- The magnitude of the effect is uncertain, however.

For example, while people might like to work more if tax rates are lowered, they might be constrained by employers expecting a particular work week (like 40 hours).

Economists believe that more study is needed on the impact of tax reforms.



Appendix A Closer Look at the Multiplier Formula

Learning Objective

Apply the multiplier formula.

Our objective in this appendix is to develop an *econometric* model for how real GDP is determined.

Then we will use that model to identify:

1. The government purchases and tax multipliers
2. How those multipliers are altered by tax rates
3. How those multipliers change in an *open economy*, i.e. when net exports change in response to income changes.

Throughout, we will assume that price level is constant.

An Expression for Equilibrium Real GDP (1 of 2)

For simplicity, we will initially assume that taxes do not depend on income (i.e. they are a fixed amount), there are no government transfers to households, and there are no imports or exports.

1. $C = 1,000 + 0.8(Y - T)$	Consumption function
2. $I = 125$	Planned investment function
3. $G = 125$	Government purchases function
4. $T = 100$	Tax function
5. $Y = C + I + G$	Equilibrium condition

These numbers are in billions of dollars, except the marginal propensity to consume (MPC) of 0.8.

Consumption depends on disposable income $Y - T$.

Then we can write:

$$Y = 100 + 0.8(Y - 100) + 125 + 125$$

An Expression for Equilibrium Real GDP (2 of 2)

$$\begin{aligned} Y &= 100 + 0.8(Y - 100) + 125 + 125 \\ &= 100 + 0.8Y - 80 + 125 + 125 \end{aligned}$$

Subtracting $0.8Y$ from both sides, we obtain:

$$\begin{aligned} Y - 0.8Y &= 100 - 80 + 125 + 125 \\ 0.2Y &= 270 \end{aligned}$$

$$Y = \frac{270}{0.2} = 1350$$

Thus we conclude that according to our model, real GDP will be \$13 trillion.

An Algebraic Expression for Equilibrium Real GDP

Instead of inserting numbers, we could find an expression for real GDP algebraically. Let:

$$C = \bar{C} + MPC(Y - \bar{T})$$

Consumption function

$$I = \bar{I}$$

Planned investment function

$$G = \bar{G}$$

Government purchases function

$$T = \bar{T}$$

Tax function

$$Y = C + I + G$$

Equilibrium condition

(The letters with bars represent fixed or *autonomous* variables.)

Then:

$$Y = \bar{C} + MPC(Y - \bar{T}) + \bar{I} + \bar{G}$$

or

$$Y - MPC(Y) = \bar{C} - (MPC \cdot \bar{T}) + \bar{I} + \bar{G}$$

or

$$Y(1 - MPC) = \bar{C} - (MPC \cdot \bar{T}) + \bar{I} + \bar{G}$$

or

$$Y = \frac{\bar{C} - (MPC \cdot \bar{T}) + \bar{I} + \bar{G}}{1 - MPC}$$

A Formula for the Government Purchases Multiplier

Any change in an autonomous variable will change the value of real GDP (Y). We can make that explicit by writing the variables in *change form*:

$$\Delta Y = \frac{\Delta \bar{C} - (MPC \times \Delta \bar{T}) + \Delta \bar{I} + \Delta \bar{G}}{1 - MPC}$$

This uses the standard notation of Δ for “change”.

Keeping autonomous consumption (C), taxes (T), and investment (I) constant, we get:

$$\Delta Y = \frac{\Delta G}{1 - MPC}$$

Divide both sides by ΔG to obtain the *government purchases multiplier*:

$$\frac{\Delta Y}{\Delta G} = \frac{1}{1 - MPC}$$

For example, if $MPC = 0.8$, the government purchases multiplier will be:

$$\frac{1}{1 - 0.8} = 5$$

A Formula for the Tax Multiplier

Any change in an autonomous variable will change the value of real GDP (Y). We can make that explicit by writing the variables in *change form*:

$$\Delta Y = \frac{\Delta \bar{C} - (MPC \times \Delta \bar{T}) + \Delta \bar{I} + \Delta \bar{G}}{1 - MPC}$$

Now we will keep autonomous consumption (C), investment (I), and government purchases (G) constant:

$$\Delta Y = \frac{-MPC \times \Delta T}{1 - MPC}$$

Divide both sides by ΔT to obtain the *tax multiplier*:

$$\frac{\Delta Y}{\Delta T} = \frac{-MPC}{1 - MPC}$$

For example, if $MPC = 0.8$, the tax multiplier will be:

$$\frac{-0.8}{1 - 0.8} = -4$$

A \$10 billion increase in taxes decreases real GDP by \$40 billion.

The “Balanced Budget” Multiplier

Suppose we increase government spending and taxes both by \$10 billion. What would happen to real GDP?

Increase in real GDP from the increase in government purchases = \$10 billion $\times \frac{1}{1-MPC}$

Decrease in real GDP from the increase in taxes = \$10 billion $\times \frac{-MPC}{1-MPC}$

Adding these effects and factoring out the \$10 billion gives:

$$\$10 \text{ Billion} \times \left\{ \frac{1}{1-MPC} + \frac{-MPC}{1-MPC} \right\}$$

We can simplify this to:

$$\$10 \text{ Billion} \times \left\{ \frac{1-MPC}{1-MPC} \right\} = \$10 \text{ Billion}$$

So the balanced budget multiplier is 1: equal dollar increases in government spending and taxes lead to the same dollar increase in real GDP in the short run.